Graph Databases as a Psychological Method

Juan C. Correa[1*] and María del Pilar García-Chitiva[2]

[1*]Departmento de Estudios Empresariales, Universidad Iberoamericana, Mexico City, Mexico, 01219.

[2]Escuela de Educación, Politécnico Grancolombiano, Bogotá, Colombia, 110231.

*Corresponding author(s). E-mail(s): juanc.correa@ibero.mx;

Abstract

This article provides a tutorial on labeled property graphs as a graph database paradigm well suited for psychological research. We show how this paradigm addresses the tabular constraints that fragment, duplicate, or obscure the relational structure of human behavior, and how it supports the representation, management, and analysis of complex behavioral data. Using the Speed-Dating replication dataset as a running example, we demonstrate that graph databases offer three methodological advantages particularly relevant to psychological research: (a) schema flexibility that accommodates evolving theoretical models, (b) query performance that exploits local relational structure rather than global table joins, and (c) ontological transparency that makes assumptions about entities and relationships explicit and testable.

Keywords: Graph database, Network modeling, complex behavior

1 Introduction

The relational nature of psychological data is evident across multiple research domains. For example, Guimerà, Uzzi, Spiro, and Amaral (2005) examined team assembly mechanisms to show how patterns of collaboration emerge from relational constraints and Borsboom and Cramer (2013) introduced network-based models to analyze the causal interplay among psychopathological symptoms. What these approaches share is that their relational structure is treated as an analytical construction, reconstructed from rectangular data formats, rather than as a database design principle. Treating psychological data as a graph database design represents a fundamentally different methodological paradigm that has received little explicit attention in psychological research.

Relational databases dominate psychological research for good reasons. They store data in tables where columns represent variables, rows represent cases, and relationships between entities are managed through the so-called "Structured Query Language" (SQL). This paradigm works well when data are rectangular and relationships are simple (Soto, 2025). However, as psychological phenomena grow more complex (i.e., spanning multiple levels of analysis, unfolding over time, and involving entities that relate to each other in qualitatively distinct ways) the tabular constraints of relational databases become a methodological liability rather than an asset.

An alternative paradigm, broadly referred to as "Not Only SQL" (NoSQL), includes systems designed for non-tabular, highly connected data. Among these, graph databases are particularly well suited for psychological research because they treat relationships as first-class elements rather than as implicit byproducts of table joins (Robinson, Webber, & Eifrem, 2015). This database paradigm allows behavior to be modeled as a complex system in a way that differs fundamentally from traditional approaches (Guastello, Koopmans, & Pincus, 2009). In this article, we examine graph databases as a methodological paradigm in which nodes and edges (rather than tables and joins) constitute the basic representational units of behavior. Understanding why this database paradigm remains largely absent from psychological research requires first acknowledging what already exists.

Computational libraries such as `sna`, `igraph`, `qgraph`, or `psychonetrics` have made network analysis accessible to researchers working entirely within `R` (Luke, 2015), and their adoption in psychology is well-known (Mair, 2018). However, these libraries mainly address the statistical analysis of relational data, leaving the management of it under the hood (i.e., the data must first be extracted from a rectangular format, reshaped into an edge list or adjacency matrix, and then passed to the analytic tool). Thus, such a workflow reconstructs relational structure at the analysis stage rather than preserving it from the moment of data collection. Graph database management systems, by contrast, store, query, and retrieve data in graph form natively, so that relational structure is not an analytical construction but a representational commitment. To our knowledge, this distinction has received no systematic treatment as a psychological method.

This article provides a step-by-step tutorial for implementing labeled property graphs (LPGs) in psychological research. We illustrate the approach with the Speed-Dating replication dataset (Selterman, Chagnon, & Mackinnon, 2015, 2023), a publicly available resource whose dyadic, longitudinal, and multimodal structure provides an ideal scenario for demonstrating how LPGs preserve relational topology, handle structural missingness without numerical encoding, and support queries that rectangular formats cannot express without costly joins or data duplication. The tutorial covers the full workflow: designing a graph schema, importing data into Neo4j, querying the graph with Cypher, connecting Neo4j to R via the neo2R package, and conducting statistical analysis and visualization using `tidygraph` and `ggraph`. All materials — including annotated Cypher and R code, data, and a reproducible RMarkdown file — are available in a public repository.

The remainder of this article is organized as follows. Section 2 introduces the conceptual foundations of graph databases, contrasting their representational logic with that of relational databases and defining the core elements — nodes, edges, labels, and properties — that make them particularly well-suited for psychological data. Section 3 motivates our tutorial through the Speed-Dating replication dataset (Selterman et al., 2023). This tutorial shows how this dataset and its dyadic, longitudinal, and multimodal structure exposes the limitations of rectangular formats. Section 4 presents the step-by-step tutorial itself, covering the full workflow from graph schema design and data import in `Neo4j` through querying with `Cypher` and statistical analysis and visualization in R via the neo2R, tidygraph, and ggraph packages. Section 5 discusses the broader methodological implications of graph databases for psychological research, with particular attention to structural missingness, scalability, and ontological transparency. Section 6 concludes with a discussion of limitations and directions for future work. All code, data, and a fully reproducible RMarkdown file are available via the Open Science Framework at [OSF link].

2 Graph Databases and Labeled Property Graphs

Graph databases are data management systems that implement the basic operations of **C**reate, **R**ead, **U**pdate, and **D**elete (CRUD) on a graph data model, in which entities are represented as nodes and relationships are represented as edges (Robinson et al., 2015). A fundamental element of this paradigm is the labeled property graph (LPG), in which both nodes and edges can carry labels and properties, allowing attributes of entities and relationships to be stored directly within the graph structure. Unlike relational databases (i.e., where relationships are encoded indirectly through foreign keys and join operations) graph databases treat relationships as first-class components of the data model. As a result, relational structure becomes an explicit feature of data representation, rather than a construct introduced retrospectively at the analytical stage of research.

The graph database paradigm maps naturally onto relational phenomena for psychological researchers. Participants, events, instruments, or observational units can be represented as nodes, whereas dyadic interactions, temporal transitions, or participation in shared activities can be represented as edges. Figure 1 illustrates this representation using the Speed-Dating dataset (Selterman et al., 2023), in which participants and dating events are modeled as nodes connected by relational ties derived from observed interactions.

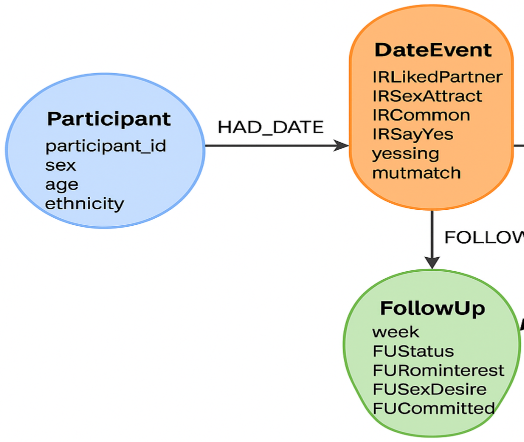

Fig. 1 A labeled property graph of the speed-dating dataset by Selterman et al. (2023)

The labeled property graph illustrated in Figure 1 helps to recreate the context of the study published by Eastwick and Finkel (2008) and replicated by Selterman et al. (2023). Here, this design enables researchers to traverse, query, and model highly interconnected data structures with exceptional efficiency and originality compared with what a researcher could do from the traditional relational database paradigm. Figure 1 shows the data at three distinct levels: (1) stable participant characteristics (e.g., age, ethnicity, sex), (2) interaction-level impressions recorded during each brief date (e.g., attraction, perceived similarity, desire to say "yes"), and (3) longitudinal follow-up responses across subsequent weeks for pairs who continued interacting. These layers of data are represented as interconnected components of a single, coherent graph structure. On this basis, the following sections describe how psychological entities and relationships can be instantiated explicitly within an LPG.

2.1 Entities as Nodes

`Participant` nodes represent individuals in the study and store stable demographic properties such as age, ethnicity, and sex (from the codebook variables DemAge, DemEthn, and sex_M1_F2). The `DateEvent` node encompasses a series of dyadic interactions and store the full set of impression-formation ratings provided by participants during speed-dating (e.g., IRLikedPartner, IRSexAttract, IRCommon, IRSayYes, yessing, mutmatch). These correspond directly to the interaction-record variables in the dataset. The `FollowUp` node represents week-level longitudinal responses, including relationship status, romantic interest, sexual desire, and commitment indicators (e.g., FUStatus, FURomInterest, FUSexDesire, FUCommitted), captured only for participants who matched and continued interacting.

2.2 Relationships as Edges

Edges define how these nodes are linked and encode psychological structure directly. The HAD_DATE edge connect participants to the DateEvent in which they took part, indicating the rater's perspective on that interaction. The edge labeled as WITH connects the DateEvent back to the other participant, allowing the system to explicitly represent dyadic structure. The edge labeled as FOLLOWED_UP links Participants to their FollowUp node, representing relational trajectories across weeks. Optionally, a MUTUAL_MATCH_WITH edge can be created between two Participants when the mutmatch property equals 1, providing a direct representation of reciprocal liking (i.e., a relationship that is invisible in rectangular data but naturally expressed in graph form).

2.3 Why Graph Databases Matter for Psychological Research

The structure of Figure 1 offers several methodological advantages. Psychological data retain their natural relational structure. Dyadic interactions are modeled as shared events rather than duplicated rows, preserving the topology of social and romantic processes as they occur in the real world. Temporal processes become navigable. The FollowUp node can be traversed sequentially, enabling queries such as "How early do committed feelings emerge among mutually matched pairs?' ' or "Does initial sexual attraction predict later romantic interest?' '

In addition, missingness is handled structurally, not numerically. Participants who did not match simply lack FollowUp; no NA encoding is required. The model is inherently extensible. Additional constructs, such as personality traits or behavioral logs, can be attached as new nodes or properties without reconfiguring the existing database schema. In sum, LPGs allow behavioral scientists to represent complex, multi-level psychological phenomena in a way that relational tables cannot. The Speed-Dating schema illustrated in Figure 1 makes clear how impression formation, dyadic reciprocity, and longitudinal relationship development can be captured and analyzed within a single unified model.

3 Graph Databases: Epistemic Considerations

In this section, we focus not on implementation, but on the epistemic considerations of adopting graph databases in psychological research. Graph databases and their labeled property graph paradigm provide an infrastructure that enables the systematic incorporation of complex systems thinking into psychological inquiry. In psychology and the behavioral sciences, complex systems thinking is not novel (Guastello et al., 2009; Luce, 1999), although more recent work has emphasized its interdisciplinary potential (Krakauer, 2019). This emphasis aligns with recent reviews that propose methodological innovations as a means of strengthening intra- and interdisciplinary integration across basic and applied domains of behavior analysis (Elcoro, Diller, & Correa, 2023).

As a psychological method, the graph database paradigm shares some similarities with the *nomological nets* originally proposed by (Cronbach & Meehl, 1955). Both emphasize relational structure and the systematic articulation of theoretical commitments. However, labeled property graph models constitute a more general-purpose representational framework that transcends the scope of classical nomological nets (see Table 1).

Table 1: Knowledge graphs and nomological nets

Feature	Labeled Property Graphs	Nomological nets
Primary field	Computer science / AI / Data engineering	Psychology / Philosophy of Science
Primary goal	Data integration, search, automated reasoning	Establishing construct validity (i.e., theory testing)
Nature of nodes	Concrete entities (e.g., “Lee J. Cronbach”, “University of Illinois”)	Theoretical constructs (e.g., “introversion”, “intelligence”)
Nature of edges	Predicates or properties (e.g., “works_at”, “collaborates_with”)	Theoretical laws or empirical correlations
Verification	Data consistency and retrieval accuracy	Statistical evidence (e.g., correlations, p-values)
Scale	Potentially billions of facts	Typically small, theory-specific

What distinguishes graph databases from analytic tools is that they constrain and enable psychological inquiry at the level of *representation*. Whereas analytic methods operate on data that have already been structured, graph databases determine how behavioral phenomena are encoded from the outset, thereby delimiting which relations, processes, and queries are epistemically viable.

This representational commitment has direct methodological consequences, particularly for research involving complex, relational, and large-scale psychological data. One such consequence concerns scalability. Because graph databases organize information around localized connections among nodes and edges, they maintain relatively stable query performance even in datasets comprising millions of entities and relationships. From a traditional perspective within psychology, working at this scale may appear implausible. However, large-scale behavioral data are now firmly within methodological consideration, as reflected in recent treatments of big data as a psychological method (Vezzoli & Zogmaister, 2023). Prior work in network science illustrates this point clearly. For example, studies of urban mobility have examined individual-level behavioral trajectories by tracking the movements of hundreds of thousands of drivers at fine temporal resolutions (Gonzalez, Hidalgo, & Barabasi, 2008). In such contexts, graph databases are particularly well suited because queries operate on localized subgraphs rather than requiring exhaustive scans of entire datasets. Performance, therefore, is not merely a technical concern but a condition for the epistemic viability of certain classes of psychological questions.

A second epistemically consequential feature of graph databases is their flexibility. Behavioral scientists routinely connect data in ways that evolve alongside theoretical commitments, empirical discoveries, and domain expertise. For this reason, graph databases often serve as the underlying infrastructure for constructing *knowledge graphs* (Barrasa & Webber, 2023). Unlike rigid schemas defined in advance, knowledge graphs allow data models to evolve in tandem with scientific understanding—an especially important feature in domains characterized by limited theory, active replication efforts, or rapid evidence accumulation (Burgos, 2025). Because graph structures are inherently additive, new nodes, relationships, labels, and subgraphs can be introduced without invalidating existing representations. From a methodological perspective, such modifications should be understood not as ad hoc adjustments, but as legitimate indicators of scientific progress under conditions of incomplete knowledge.

These representational choices bring ontological and epistemological considerations to the foreground. Ontologically, graph databases require researchers to specify explicitly which entities exist within the model, how they relate to one another, and which transformations of those entities are permissible. Epistemologically, they invite reflection on what counts as evidence, which queries are meaningful, and how explanations are warranted given the structure of the graph. In this sense, graph databases make explicit assumptions that often remain implicit in rectangular data workflows, echoing recent calls in theoretical and philosophical psychology to treat ontology and epistemology as integral components of methodological design (Burgos, 2025).

These considerations are particularly salient in contemporary psychological research that relies on automated data collection techniques. Mobile applications, such as ecological momentary assessment tools, prompt participants to report emotions or behaviors repeatedly throughout the day, while motion sensors and radio-frequency identification tags continuously record movement and location in naturalistic environments (Bak et al., 2021). In such technology-mediated inquiries, datasets tend to grow rapidly while becoming increasingly heterogeneous. Although Soto (2025) strongly advocates for relational databases in behavioral research, he also identifies several practical challenges to their adoption, including infrastructural requirements and specialized technical expertise. From the perspective advanced here, these challenges highlight the benefits of alternative data representation paradigms—such as graph databases—which align representational structure more closely with the relational nature of psychological phenomena.

In sum, labeled property graphs are not merely a technical alternative to relational schemas; they represent a methodological advance that aligns with psychology's need for flexible, scalable, and conceptually transparent data models.

4 Applying graph database in behavioral research

This section provides a step-by-step tutorial demonstrating how to (a) design a graph schema for psychological data, (b) import data into a labeled property graph, (c) query relational and longitudinal structures that are difficult to express in rectangular formats, and (d) interface the graph with R for statistical analysis and visualization.

4.1 STEP 1: Prerrequisites and materials

This tutorial examines a dataset on men and women exhibited preferences for mates (Selterman et al., 2015, 2023) which is a replication work from the original study published by Eastwick and Finkel (2008). This dataset is available in an Open Science Framework repository. This repo provides access to several files. In the files preview menu, the `speed.dating.2.zip` can be downloaded from the Final Report option. The zip file contains the raw data in an SPSS data format (.sav). This file can be easily read in SPSS, PSPP, Jasp, Jamovi, or RStudio. We decided to use RStudio to open the original dataset.

Given the wide variety of graph database management systems in the market (e.g., Neo4j, Microsoft Azure Cosmos, Aerospike, Amazon Web Services Neptune, NebulaGraph, Memgraph, TigerGraph, Giraph), the rest of this work relies on RStudio and Neo4j (Barrasa & Webber, 2023). While RStudio is well-known in psychology, Neo4j might require a brief presentation. Neo4j ranks as a leading freemium software for graph database management systems and has been used in several industry sectors including retail, finance, pharmaceutics, hospitality, and electronic commerce, among others. Neo4j offers free and enterprise editions, which can be installed on local or server environments following official documentation. As the official documentation provides helpful material for newcomers, downloading and installation details for Mac, Windows, and Linux users are not necessary here, so the reader can consult specific details elsewhere (Van Bruggen, 2014).

4.2 STEP 2: Instance creation and data import

As Neo4j does not support SPSS datasets, we suggest to open the .sav file in RStudio with the standard `read_sav` syntax provided by the `haven` R library (Wickham, Miller, & Smith, 2025) and exported it as a standard coma separated value file (.csv) with the `write.csv` syntax provided by the `utils` R library (R Core Team, 2025). We named this csv file as `speed_dating.csv`.

4.3 STEP 2: Understanding the context

4.4 STEP 3: Sketching a graph database model

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